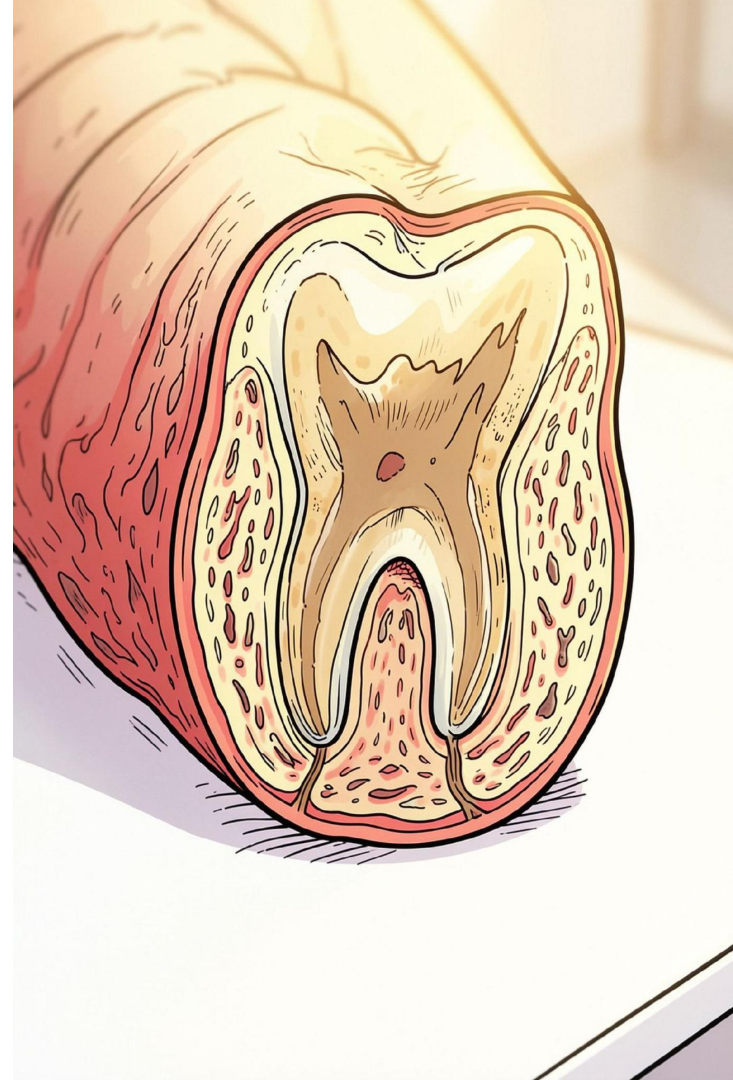


Cementum

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Learning Objectives

1 **Composition & Structure**
Know the composition and main structural features of cementum

2 **Types of Cementum**
Be familiar with the various types and classifications of cementum

3 **Key Junctions**
Appreciate the importance of cementum–enamel and cementum–dentinal junctions

4 **Cementum vs. Bone**
Compare and contrast cementum with bone tissue

5 **Clinical Relevance**
Understand clinical importance during orthodontic movement and periodontal disease

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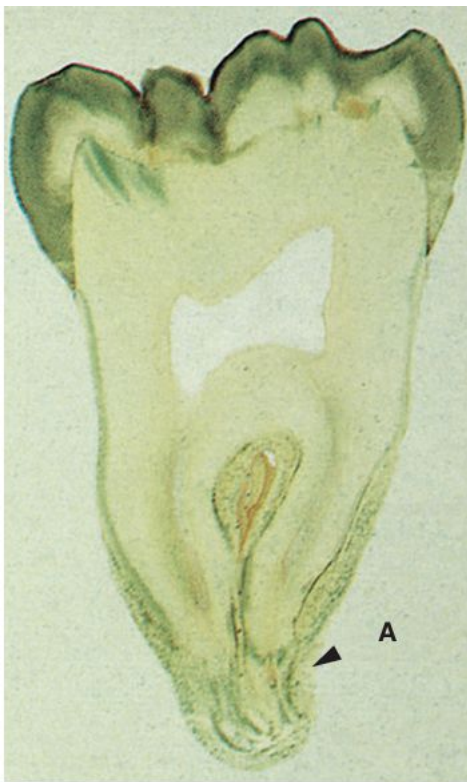
Functions of Cementum

Introduction

Cementum is a **thin layer of calcified tissue** covering the dentine of the root, anchoring the tooth to the surrounding alveolar bone via the periodontal ligament (PDL).

- Lines the root surface; forms two critical junctions with enamel and dentine
- Composition similar to bone but **lacks blood vessels and nerves**
- Organic matrix composed primarily of **Type I collagen**
- Slowly deposited throughout life
- Derived from the investing layer of the **dental follicle**





Physical & Chemical Properties

Hardness & Color

Less hard than dentine; light yellow in color and lacks the luster of enamel

Thickness Variation

Thinnest at the cemento-enamel junction (**10–15 μm**); thickest at root apex and interradicular areas (**50–200 μm** , may exceed 600 μm)

Prime Function

Attachment of collagen fibers of the periodontal ligament; maintains root integrity and tooth position

Inorganic (Dry Weight)

45–50% — hydroxyapatite crystals (calcium & phosphate); platelike and less mineralized than enamel

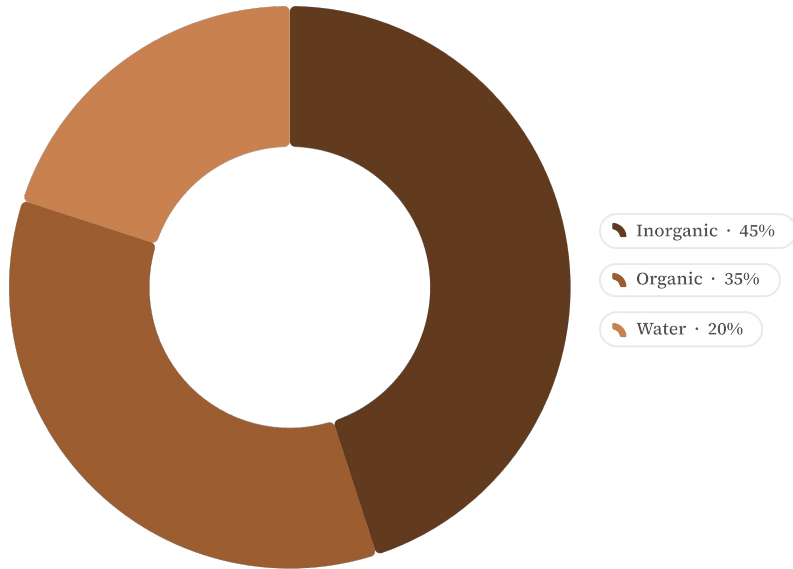
Organic Matrix

50–55% — Type I collagen (90%), Type III collagen (5%), non-collagenous proteins

Water

~20% by volume — rich in glycosaminoglycans; fluoride levels higher than in enamel and dentine

Composition Details



Key Compositional Highlights

- Principal inorganic component: **hydroxyapatite** — thin, platelike crystals
- Less mineralized overall than enamel and dentine
- Rich in **glycosaminoglycans**
- Fluoride levels **higher** than in enamel and dentine
- Non-collagenous proteins: **Sialoproteins** (matrix) and **Osteopontin** (incremental lines)
- Growth factors present — activated when required for repair

Composition Details

- The organic matrix is primarily collagen.
- The collagen is virtually all type I (although types III, V, VI and XII have been found in small quantities).
- Molecules known to be present are bone sialoprotein, dentine sialoprotein, fibronectin, tenascin and osteopontin and possibly other cementum-specific elements that are conjectured to be involved in periodontal reattachment.
- Sialoproteins are located in the cementum matrix and osteopontin in the incremental lines
- Cementum-derived attachment protein (CAP) is a 56 kDa or 65 kDa collagenous protein in cementum that promotes the attachment of mesenchymal cells to the extracellular matrix. It is located in the matrix of mature cementum and in cementoblasts (but not in bone). Thus, CAP may be a marker to differentiate bone and cementum.

Composition Details

- Cementum is rich in glycosaminoglycans, predominantly chondroitin sulphate.
- This glycosaminoglycan is primarily located around the lacunae of cementum.
- In addition, dermatan sulphate and hyaluronan can be found.
- Among the proteoglycans found in cementum are lumican, versican, decorin, biglycan and fibromodulin.
- These are also found in bone and, for both tissues, are located especially at the peripheries of lacunae and canaliculi.
- It appears that fibromodulin and lumican are more abundant in cementum.

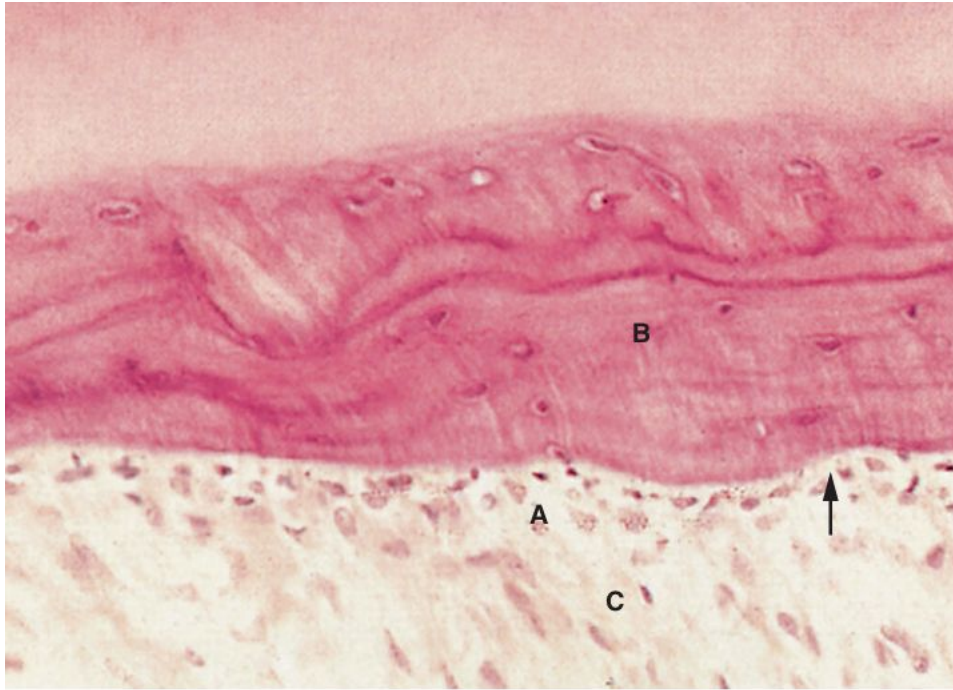


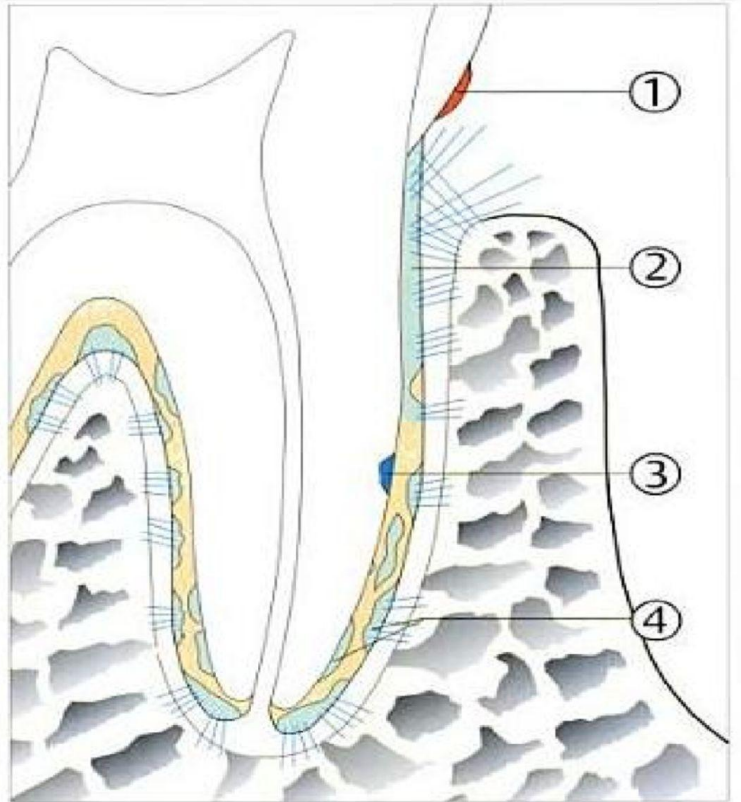
fig. 11.2 The relationship between cementum (B), precementum (arrow), a layer of cementoblasts (A) and the periodontal ligament (C) (decalcified section; H & E; $\times 200$).

There is always a thin layer (3–5 μm) of uncalcified matrix on the surface of the cellular variety of cementum (see page 414). This layer of uncalcified matrix is called **PRECEMENTUM** **OR CEMENTOID**

Classification: Acellular vs. Cellular Cementum

26 Types of Cementum— Structure, Localization and Development

- 1 **Acellular, A fibrillar Cementum** (AAC; red) AAC is formed at the most cervical enamel border following completion of pre-eruptive enamel maturation, and sometimes also during tooth eruption. It is probably secreted by cementoblasts.
- 2 **Acellular, Extrinsic-fiber Cementum** (AEC; green) AEC forms both pre- and post-eruptively. It is secreted by fibroblasts. On the apical portions of the root, it comprises a portion of the mixed-fiber cementum.
- 3 **Cellular, Intrinsic-fiber Cementum** (CIC; blue) CIC is formed both pre- and post-eruptively. It is synthesized by cementoblasts, but does not contain extrinsic Sharpey's fibers.
- 4 **Cellular, Mixed-fiber Cementum** (CMC; orange/green) CMC is formed by both cementoblasts and fibroblasts; it is a combination of cellular intrinsic-fiber cementum and acellular extrinsic-fiber cementum.

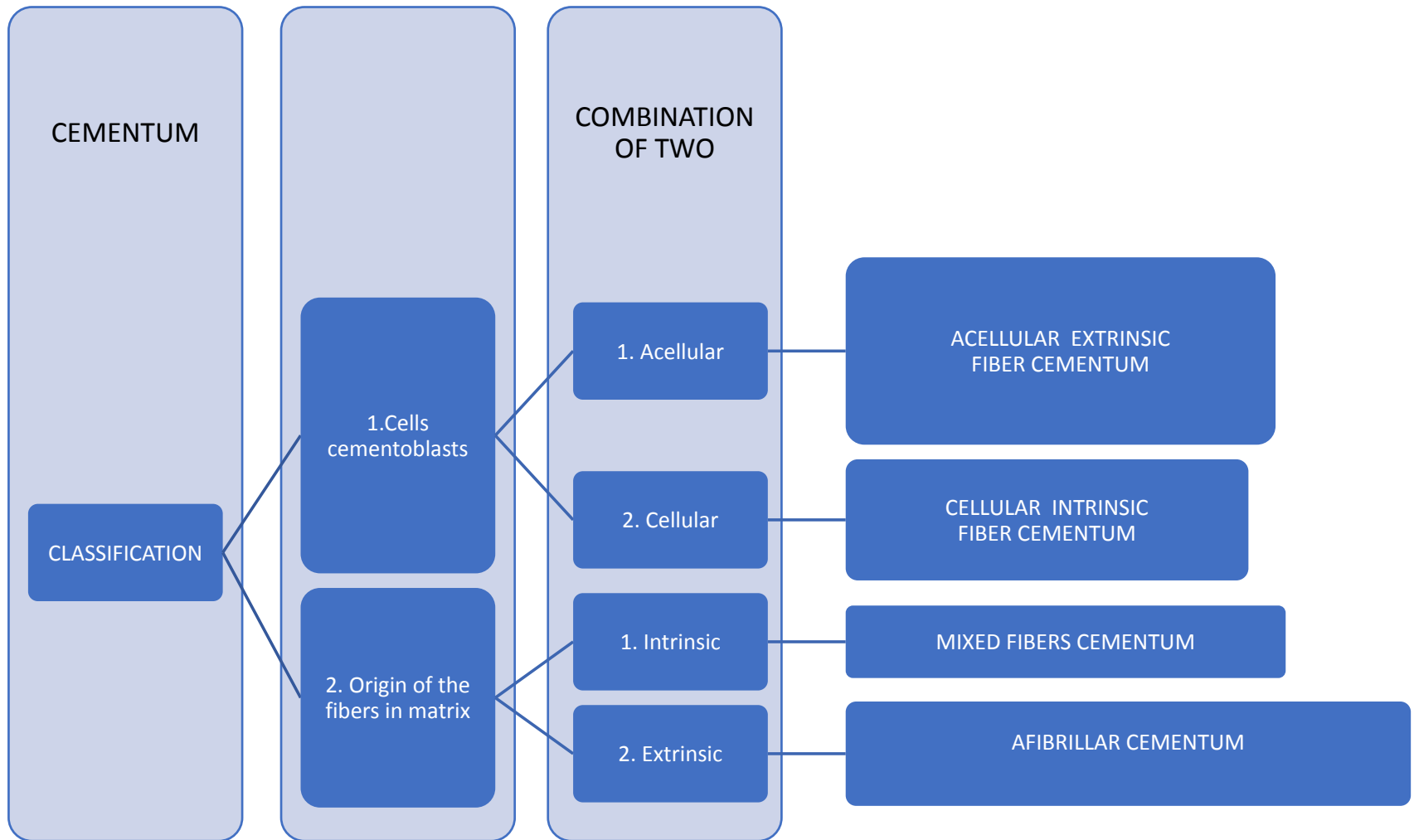


Acellular Cementum

- Covers root adjacent to dentine
- Forms during root formation — **before** tooth reaches occlusal plane
- Most common arrangement
- No embedded cells

Cellular Cementum

- Found mainly in the **apical area**, overlying acellular cementum
- Forms after eruption once tooth reaches occlusal plane
- Contains **cementocytes** embedded in lacunae



ACELLULAR & CELLULAR/ CEMENTUM

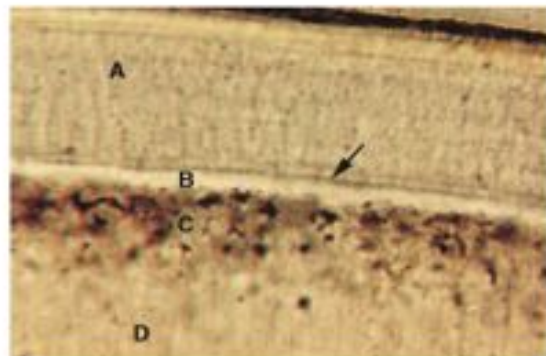


Fig. 11.7 The appearance of acellular cementum (A). B = hyaline layer (of Hopewell-Smith); C = granular layer (of Tomes); D = root dentine. Note that the dark layer arrowed between the hyaline layer and the acellular cementum may be related to the alibrillar cementum patchily present at this position (ground section; $\times 200$).



Fig. 11.8 Cellular cementum (B) overlying acellular cementum (A). Note the greater thickness of the cellular layer (ground section; $\times 50$).

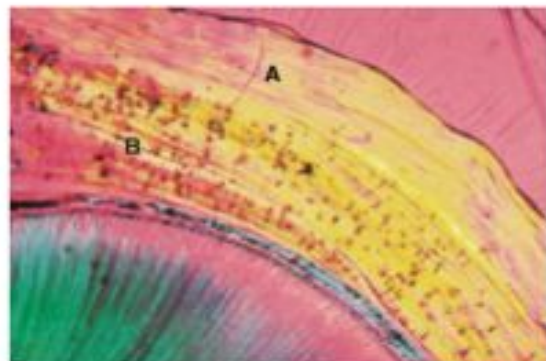


Fig. 11.9 Acellular cementum (A) overlying cellular cementum (B) (ground section of the root; $\times 50$).

In the most common arrangement,

acellular cementum covers the root adjacent to the dentine(A)

Forms during root formation before tooth reaches occlusal plane

cellular cementum is found mainly in the apical area and overlying the acellular cementum(B)

Forms after the eruption of tooth once it reaches occlusal plane

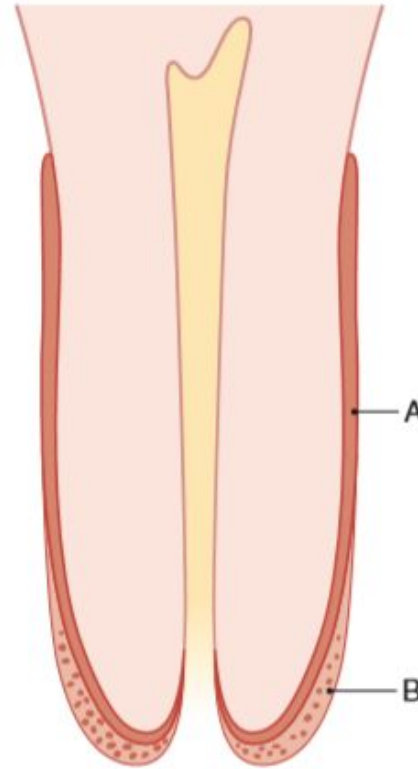
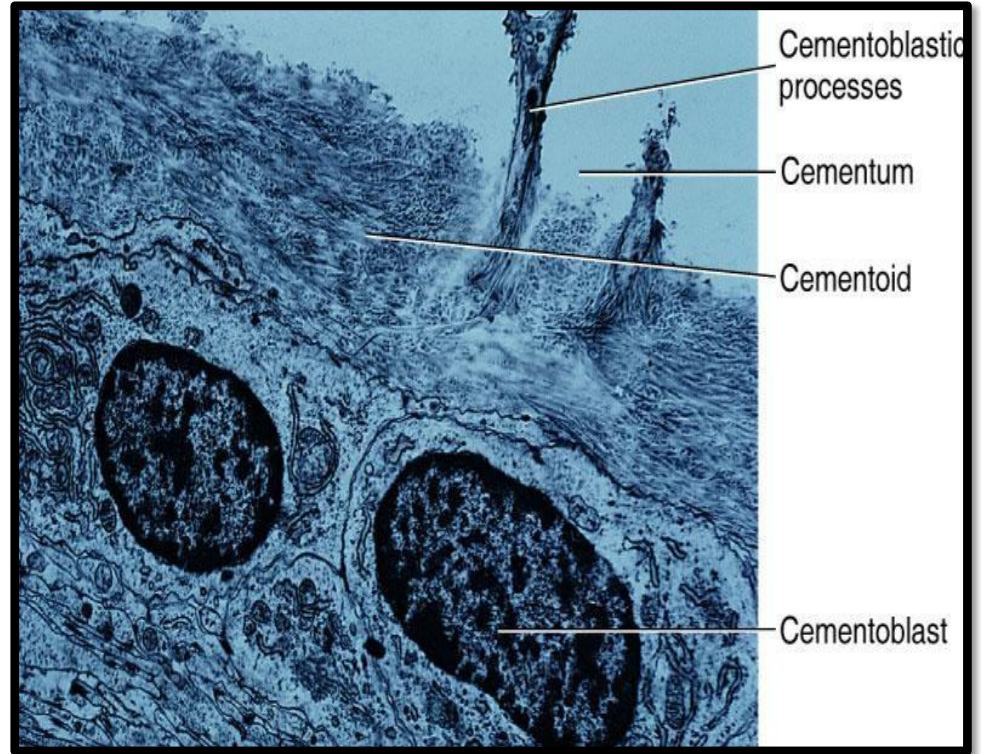
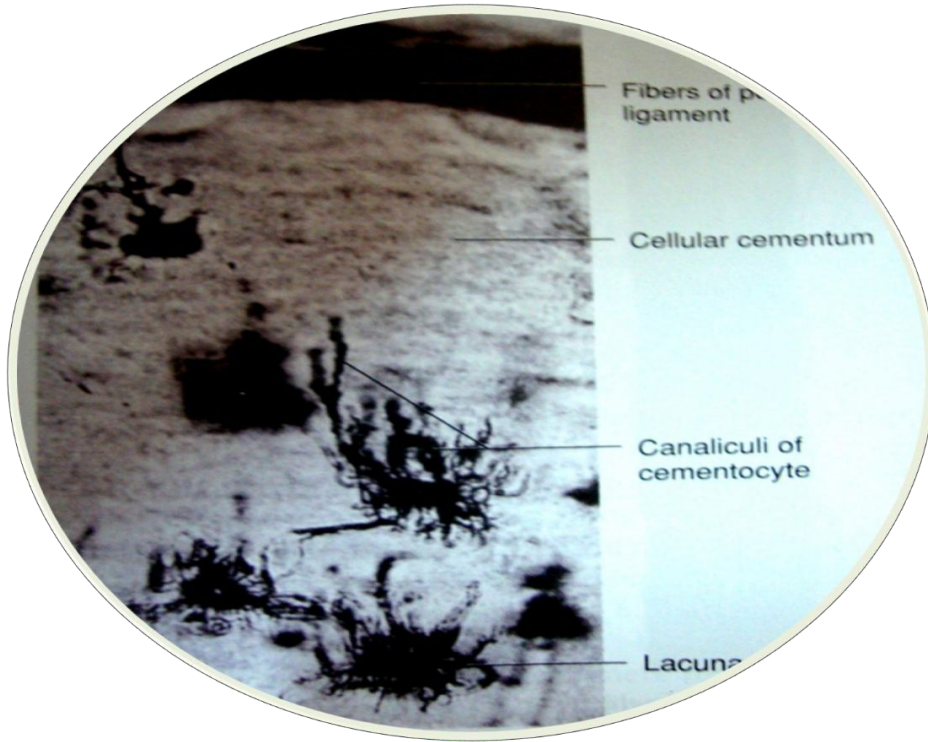


Fig. 11.6 The distribution of acellular (A) and cellular (B) cementum.

CEMENTOBLAST/CEMENTOCYTES

- ❑ Derived from dental follicle.
- ❑ Transformation of mesenchymal cells of dental follicle.
- ❑ Cementocytes → lacunae and the channels that their processes extend along are the canaliculi.
- ❑ Canaliculi facing to PDL = Nutrition
- ❑ Inactive Cementocytes





Cementoblasts & Cementocytes

→ Cementoblasts

Derived from dental follicle mesenchymal cells; resemble osteoblasts; responsible for forming cementum matrix

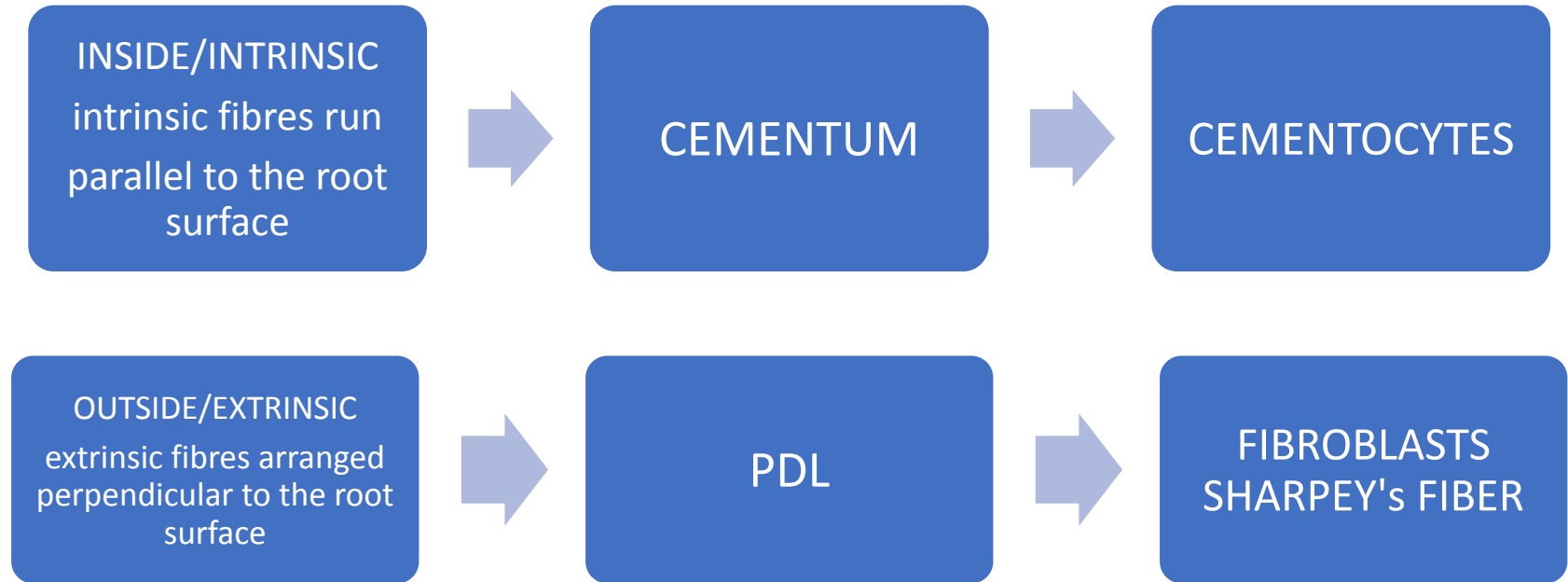
→ Cementocytes

Inactive cementoblasts embedded in lacunae; extend processes along canaliculi; communicate through anastomosing canaliculi system

→ Nutrition

Canaliculi facing the PDL provide essential nutrition to embedded cementocytes

FIBERS OF CEMENTUM



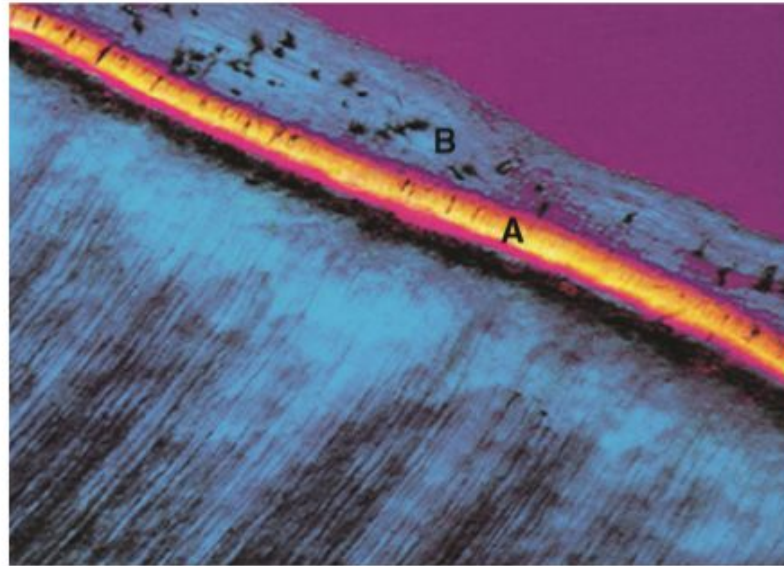


Fig. 11.20 Fibre orientation in acellular and cellular cementum. The root surface is seen in polarised light, the different colours reflecting different orientations of the collagen fibres. A = acellular cementum; B = cellular cementum (ground, longitudinal section; $\times 50$).

Two main sources of collagen fibers

- Sharpey's fibers (Extrinsic) are the embedded portion of the principal fibers of periodontal ligament and formed by fibroblasts.
- Fibers that belong to the cementum matrix (intrinsic) and produced by cementoblast.

Due to its lower crystallinity of mineral component:

- has the highest Fluoride content
- readily decalcifies in the presence of acidic conditions

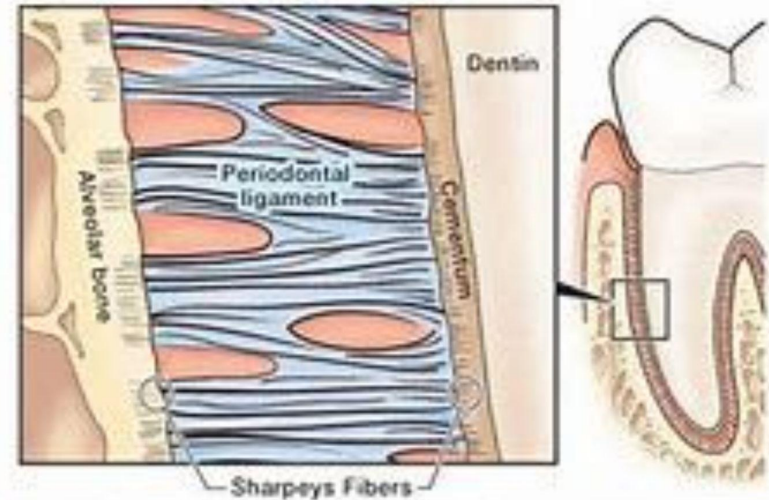


Figure 2.18. Sharpey Fibers. The ends of the periodontal ligament fibers that are embedded in the alveolar bone and the cementum are known as Sharpey Fibers.

Fibers of Cementum

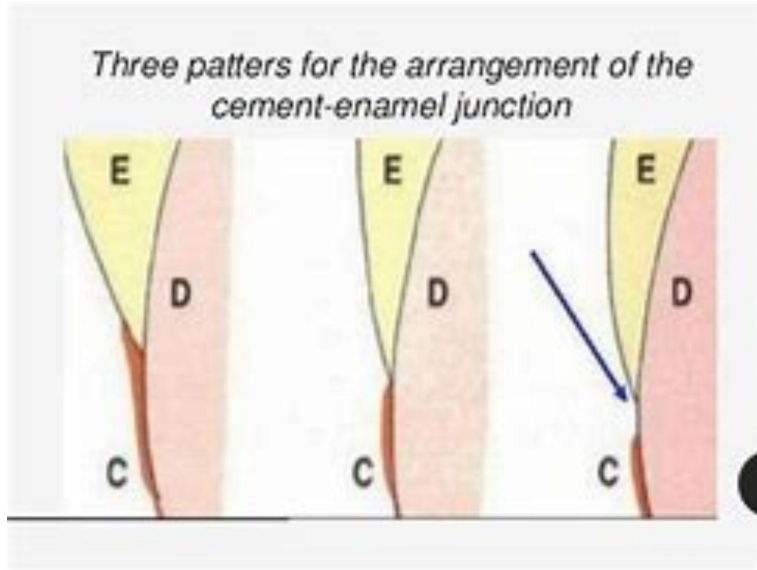
Sharpey's Fibers (Extrinsic)

- Embedded portion of **principal fibers of the PDL**
- Formed by fibroblasts
- Provide mechanical attachment to alveolar bone

Intrinsic Fibers

- Formed by **cementoblasts**
- Originate within the cementum itself
- Constitute the internal scaffolding of the matrix

These two distinct sources of collagen fibers together constitute the complete fibrous architecture of cementum.



Cementoenamel Junction (CEJ)

The CEJ is the boundary between cementum and enamel at the cervical region — a critical anatomical and clinical landmark.

Three Possible Arrangements

⚠ Exposed CEJ is a key indicator of **gingival recession** and guides periodontal assessment.

1

Overlap

Cementum overlaps enamel — **most common** arrangement

2

Edge-to-Edge

Cementum meets enamel at a precise junction

3

Gap

A gap exists between cementum and enamel — clinically significant



Clinical Significance & Key Takeaways



Attachment Integrity

Cementum anchors Sharpey's fibers of the PDL — essential for stable tooth position and withstanding occlusal forces



Orthodontic Movement

Cementum remodels in response to mechanical forces; root resorption is a recognized risk during orthodontic treatment



Periodontal Disease

Loss of cementum attachment leads to pocket formation and tooth mobility; CEJ is the reference point for clinical measurements



Repair & Regeneration

Cementum has limited but real repair capacity; growth factors within the matrix can be activated to support regeneration